SOLUTIONS ENGINEERING · TAKE-HOME DESIGN DOCUMENT

Greenlight

Automated ad compliance and brand safety for creator video ads, built on TwelveLabs

Use case: a global makeup brand promoting creator beauty videos as paid ads
Decision output: APPROVE / REVIEW / BLOCK, with an auto-generated description and timestamped evidence

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Greenlight is an automated gate that decides, for every creator video an advertiser wants to promote, whether it can run as a paid ad. It returns an APPROVE, REVIEW or BLOCK verdict, a two to five sentence description of the video, and timestamped evidence for every reason behind the call. TwelveLabs handles video perception across visuals, speech and on-screen text. A deterministic policy engine makes the verdict. This document explains how decisions are made, why the outputs are trustworthy, and how the system scales in a real ads pipeline.

1 The problem, stated as the buyer sees it

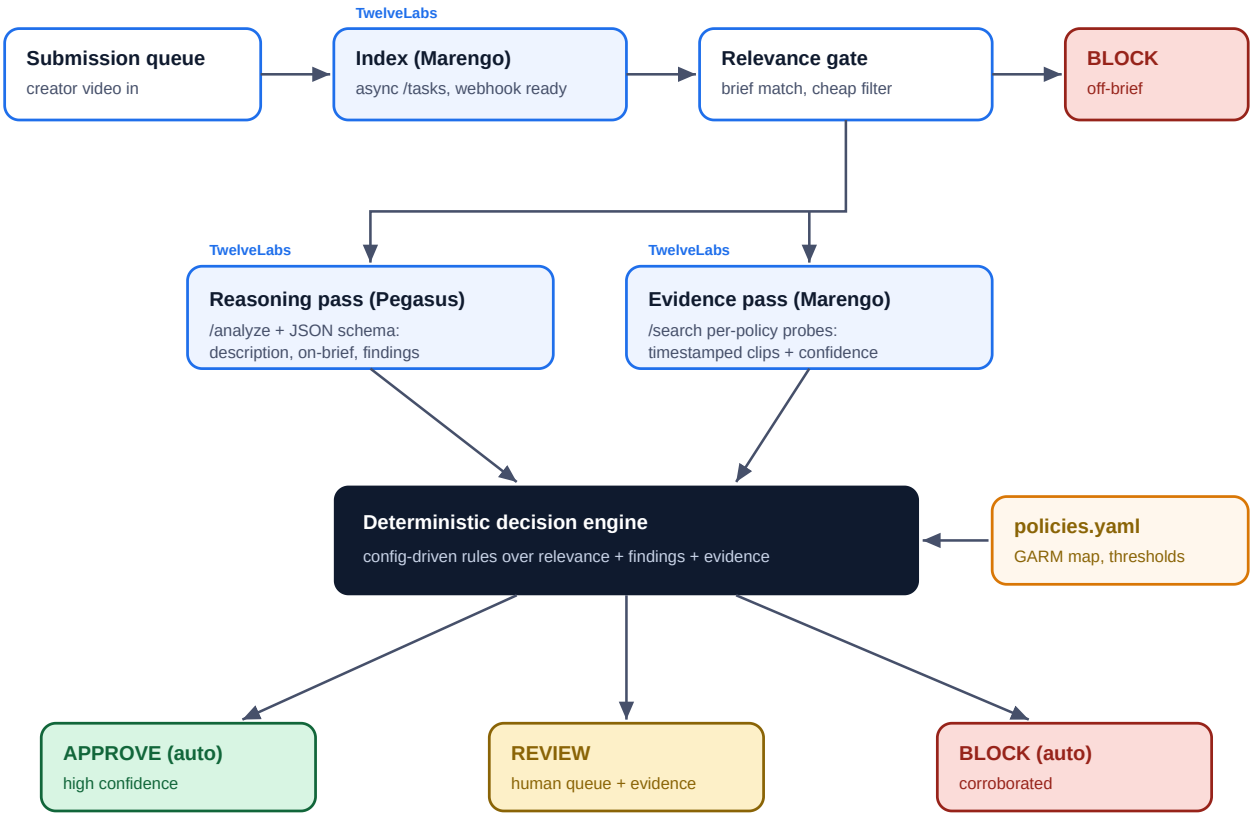
The platform's revenue depends on video advertising, and advertisers increasingly run creator-generated content as the ad creative itself. Before a beauty brand lets a creator video run as a paid ad, it expects brand safety, compliance and explainability guarantees. Today that check is manual: inconsistent between reviewers, slow, and impossible to run at the volume of submissions a large platform receives.

The hard parts are specific. Risky content can appear for only a second, in a visual, a spoken phrase, or on-screen text. Subtle violations such as unsafe product application or a medical claim dressed up as a beauty tip are easy to miss. Some submissions are off-brief: creator content that barely features the product, or unrelated clickbait. And when an ad is blocked, the advertiser and the reviewer both need to know exactly why.

> **The design thesis**
> The model perceives. The engine decides. TwelveLabs understands what happens in the video. A separate, deterministic policy engine turns that understanding into the verdict. No language model makes the final compliance call on its own. In an ads system a wrong automated block on a large advertiser is a revenue and trust incident, so the decision stays in inspectable, testable, per-advertiser-tunable rules.

2 System architecture

The pipeline runs four perception and decision stages behind an async ingestion front end. The relevance gate runs first as a cheap filter so off-brief creatives are rejected before the system spends generative compute on them.



Two independent perception passes, one deterministic decision. The reasoning pass gives an open-ended read and catches novel issues. The evidence pass corroborates specific known-bad patterns with timestamped clips. The most serious calls require both to agree.

Why two passes instead of a single prompt

A single "watch this and tell me if it is safe" call is a demo, not a system. It returns prose you cannot audit, gives no independent corroboration, and produces a decision you cannot unit-test. Splitting reasoning (Pegasus) from targeted detection (Marengo search) yields clickable timestamped evidence per policy and lets corroboration raise or lower confidence explicitly. That separation is the foundation everything else rests on.

3 Policy taxonomy

Policies live in configuration, mapped to the GARM Brand Safety Floor and Suitability Framework. Each policy declares whether it is a GARM floor category, its suitability tier, the severity at which it blocks or routes to review, and the search probes used to gather evidence. Adding a category or changing a tolerance for a new advertiser is a config edit, not a code change.

Policy	GARM	Tier	Blocks at	Reviews at
Hate, harassment or discrimination	Floor	High	medium	low
Drug use or illegal behaviour	Floor	High	medium	low
Profanity or explicit language	Suitability	Medium	high	low
Unsafe or misleading product usage	Suitability	High	medium	low
Medical or unsubstantiated cosmetic claims	Suitability	High	medium	low
Campaign relevance (on-brief)	Gate	n/a	off-brief	n/a

> **On GARM, stated honestly**
> GARM the organisation was disbanded by the World Federation of Advertisers in August 2024. Its Brand Safety Floor and Suitability Framework remain the de facto industry standard, embedded in the major verification vendors. Greenlight adopts the framework, not the defunct body. A sharp panel will know this, so the doc states it plainly.

Beauty-specific compliance: the cosmetic-versus-drug line

The most important regulatory boundary for makeup ads is the line the FDA draws between a cosmetic and a drug. A product claim crosses into drug territory when it claims to treat, cure, or affect the structure or function of the body, for example "cures acne" or "treats rosacea". Those claims are not permitted in a cosmetic ad without drug approval, so Greenlight treats them as blocking. Claims that stay in beautify, cleanse and moisturise territory are acceptable. This boundary is nuanced and a production deployment should involve counsel to maintain the claims taxonomy.

4 How the decision is made

Severity runs none, low, medium, high. The decision engine combines three inputs: the relevance score, the Pegasus findings, and the Marengo evidence. Search corroboration can lift a model finding by one severity notch, because two independent signals are stronger than one. Verdict precedence is BLOCK over REVIEW over APPROVE.

1. Off-brief (relevance below threshold and the model agrees) results in BLOCK; the creative is not eligible to run.
2. A GARM floor category, corroborated, at or above its block severity results in BLOCK.
3. Any policy at or above its block severity results in BLOCK.

4. Any policy at or above its review severity routes to human REVIEW.
5. Otherwise, APPROVE.

Confidence and human-in-the-loop. Two-source agreement is high confidence; a single source is medium. Any automated APPROVE or BLOCK below the configured minimum confidence is downgraded to REVIEW so a human looks at it. The only call permitted to auto-block at the floor is a corroborated GARM-floor violation. This is the mechanism that keeps the system from auto-acting on model uncertainty.

Worked examples from the prototype

Creative	Verdict	Why
Clean product tutorial	APPROVE	On-brief, relevance 0.88, every policy clean. High confidence.
Eye-area misuse plus medical claim	BLOCK	Unsafe usage and a drug claim, both at high severity, each corroborated by a timestamped search hit (34.0s, 57.0s).
One mild swear word, otherwise on-brief	REVIEW	Profanity at low severity, below the block threshold but above review. Routed to a human with the clip at 46.5s.
Off-brief lifestyle clickbait	BLOCK	Relevance 0.28, product only appears incidentally. Rejected at the relevance gate before deep analysis.

5 Why the outputs are trustworthy

- **Two independent signals.** The most serious automated calls require Pegasus reasoning and a Marengo search hit to agree. Independent corroboration, not a single model opinion.
- **Deterministic, testable verdicts.** Because the decision is rules over signals, it is unit-testable. The prototype ships tests proving that a given set of signals always yields the same verdict. You cannot write that test against "ask the model to decide".
- **Timestamped evidence per reason.** Every blocking or review reason carries the clip start and end and a confidence level, so a reviewer or advertiser can watch the exact moment.
- **Humans on uncertainty.** Low-confidence calls and the entire REVIEW tier route to human moderators by design, with evidence pre-attached.
- **Auditable by construction.** Each decision traces to a rule, a threshold version, and a piece of evidence, which is what an advertiser dispute six weeks later actually requires.

6 Scaling in a real ads system

- **Async indexing.** Submit to the indexing endpoint and resume on the `index.task.ready` webhook (HMAC-signed and verified). Never block a worker on a long indexing job.
- **Index once, query many.** Embeddings are cached, so repeat policy probes are cheap. Reserve the generative Pegasus pass for creatives that clear the cheap relevance pre-filter.
- **Bounded worker pool.** A queue with a worker pool sized to the concurrent-task limit handles batch pre-publish review. Pre-publish review is batch-friendly; live UGC needs tighter SLAs and a streaming-adjacent pattern.
- **Per-advertiser tenancy.** Each advertiser gets its own policy config and thresholds. Onboarding scales through configuration, not engineering.
- **Cost control as a design input.** The relevance gate exists partly to protect spend: off-brief creatives are rejected before the expensive generative pass runs.

7 Build versus buy: why TwelveLabs over a DIY stack

This is the question an evaluator will press hardest, so it is answered directly.

Dimension	DIY stack (Whisper + a vision model + an LLM)	TwelveLabs
Temporal understanding	General vision models sample roughly one frame per second and treat frames as discrete snapshots, losing temporal continuity. A one-second violation between sampled frames can be missed.	Video-native models reason over time and space directly.
What you own and maintain	ASR, frame sampling, vision inference, prompt stitching, a vector database, evaluation and ongoing model upkeep.	One platform with search and generative analysis; the team focuses on policy logic, not plumbing.

Repeat-query economics	Re-running detection means re-paying for inference each time.	Embeddings are cached, so repeat policy probes are cheap.
Evidence	You build timestamp extraction yourself.	Search hits return start, end and confidence natively.

Against hyperscaler moderation (AWS Rekognition, Google Video Intelligence, Azure Video Indexer): those excel at fixed-taxonomy primitives such as NSFW and violence with deep cloud-native integration, but offer narrower categories and less open-ended reasoning. TwelveLabs differentiates on semantic search plus generative reasoning plus custom natural-language policy without retraining. If a buyer needs only binary NSFW detection at massive scale on AWS, a Rekognition-first design may win on cost; for nuanced, explainable, custom brand suitability with semantic evidence, TwelveLabs is the stronger fit. The honest answer is a hybrid: cheap pre-filter, deep analysis where it matters, humans on the edges.

8 Honest limitations and what to verify

- The generative model can hallucinate, which is precisely why the decision is deterministic and REVIEW routes to humans.
- Indexing adds latency on long videos; the pipeline is built async to absorb it.
- Relevance-score normalisation uses an empirical ceiling that should be calibrated per index from a labelled set, not guessed.
- The cosmetic-versus-drug claim boundary is nuanced; a real deployment should involve counsel.
- The TwelveLabs API surface moves quickly. Model names, endpoints and limits should be re-verified against live documentation before relying on them; several endpoints have been renamed or retired in recent releases.

> **Integration reference used in the build**
> Marengo for multimodal embeddings, indexing and search; Pegasus for generative analysis with structured JSON output and timestamped segmentation. REST endpoints used: indexing tasks, `/analyze` with a JSON schema for the structured decision and description, and `/search` with visual, audio and transcription modalities for timestamped evidence. The prototype talks to the documented REST surface directly so it controls retries, timeouts and error handling, with a fixture-backed mock layer so the full pipeline runs in CI and in a live demo without a key. Exact model versions and endpoint paths should be confirmed in the Playground before the interview.